CHAPTER

2

Statistics, Probability and Noise

Statistics and probability are used in Digital Signal Processing to characterize signals and the processes that generate them. For example, a primary use of DSP is to reduce interference, noise, and other undesirable components in acquired data. These may be an inherent part of the signal being measured, arise from imperfections in the data acquisition system, or be introduced as an unavoidable byproduct of some DSP operation. Statistics and probability allow these disruptive features to be measured and classified, the first step in developing strategies to remove the offending components. This chapter introduces the most important concepts in statistics and probability, with emphasis on how they apply to acquired signals.

Signal and Graph Terminology

A *signal* is a description of how one parameter is related to another parameter. For example, the most common type of signal in analog electronics is a *voltage* that varies with *time*. Since both parameters can assume a continuous range of values, we will call this a **continuous signal**. In comparison, passing this signal through an analog-to-digital converter forces each of the two parameters to be *quantized*. For instance, imagine the conversion being done with 12 bits at a sampling rate of 1000 samples per second. The voltage is curtailed to 4096 (2^{12}) possible binary levels, and the time is only defined at one millisecond increments. Signals formed from parameters that are quantized in this manner are said to be **discrete signals** or **digitized signals**. For the most part, continuous signals exist in nature, while discrete signals exist inside computers (although you can find exceptions to both cases). It is also possible to have signals where one parameter is continuous and the other is discrete. Since these mixed signals are quite uncommon, they do not have special names given to them, and the nature of the two parameters must be explicitly stated.

Figure 2-1 shows two discrete signals, such as might be acquired with a digital data acquisition system. The **vertical axis** may represent voltage, light